

Project Documentation

Project Summary

The Library Book Request Management System is a ServiceNow application that digitizes the book request process for college libraries. Students submit requests through a structured form, the system automatically validates and numbers each request, the librarian is notified through an automated workflow, and all activity is visualized through a dashboard.

Technical Summary

Application Scope: Library Book Request

Table Name: `u\_library\_book\_request`

Total Fields: 9

Business Rules: 2 (Auto Request Number, Date Validation)

Automation: 1 Flow Designer workflow (Email Notification)

Reports: 2 (Pie Chart by Status, List of all requests)

Dashboard: 1 (combines both reports)

Field Reference

- Student Name — String — Required
- Student Email — Email — Required
- Book Title — String — Required
- Author — String — Optional
- Issue Date — Date — Validated (cannot be in the past)
- Return Date — Date — Optional
- Request Number — String — Auto-generated, Read-only
- Status — Choice — Default: Pending
- Librarian Notes — String — Optional

App Home		Data Table And Forms					
Library Book Request		Library Book Request					
Library Book Request		Data		Forms		Flows	
		Policies and rules		Preview		Save	
Table fields		Add new field		Filter fields		Filter options	
Column label	Column name	Type	Reference	Max length	Default value	Display	Updated
Author	author	String		40		☐	2026-06-12 22:48:16
Book Title	book_title	String		40		☐	2026-06-12 22:46:58
Created	sys_created_on	Date/Time				☐	2026-06-12 22:28:57
Created by	sys_created_by	String		40		☐	2026-06-12 22:28:57
Issue Date	issue_date	Date				☐	2026-06-12 22:48:49
Librarian Notes	librarian_notes	String		40		☐	2026-06-12 22:53:36
Number	number	String		40	javascript:global.getNext...	☐	2026-06-12 22:28:58
Request Number	request_number	String		40		☐	2026-06-12 22:52:16
Return Date	return_date	Date				☐	2026-06-12 22:49:22
Status	status	Choice				☐	2026-06-12 22:53:37
Student Email	student_email	String		40		☐	2026-06-12 22:46:00

Business Rules Reference

1. Auto Generate Request Number

- Runs: Before Insert
- Purpose: Creates a unique LBR-prefixed request number for every new record

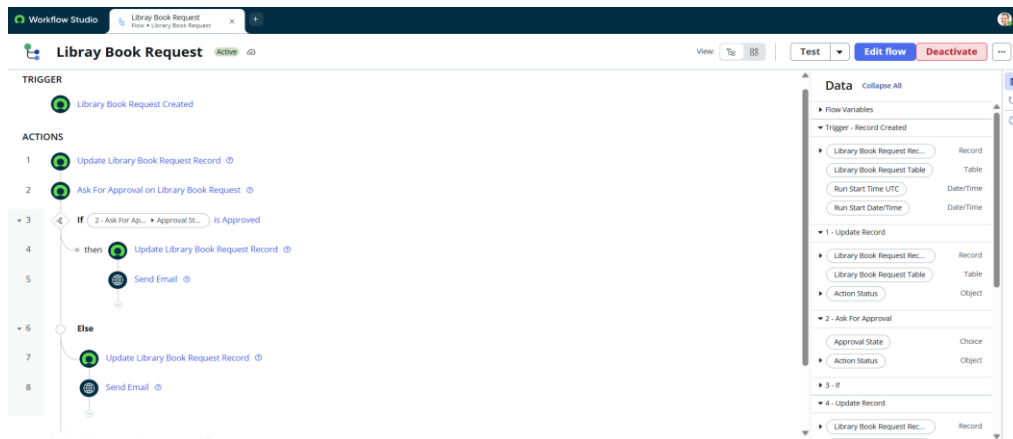
2. Validate Issue Date

- Runs: Before Insert, Before Update
- Purpose: Blocks save if the issue date is earlier than today, with a clear error message

Automation Reference

Flow Name: Library Book Request Notification

- Trigger: Record Created on `u\_library\_book\_request`
- Action: Send Email to librarian with full request details
- Verified working via Flow Designer execution logs



User Guide — For Students

1. Navigate to **Library Management** → **Library Book Request**
2. Click **New**
3. Fill in: Student Name, Student Email, Book Title, Author, Issue Date, Return Date
4. Click **Submit**
5. Note the auto-generated Request Number shown on the record for future reference
6. If the Issue Date entered is in the past, an error will appear — select today's date or a future date

User Guide — For Librarian / Admin

1. Navigate to the **Library Book Request Dashboard** for a quick summary
2. Review the Pie Chart to see request volume by status
3. Open individual requests from the list to review details
4. Update the **Status** field to Approved or Rejected as appropriate
5. Optionally add comments in the **Librarian Notes** field
6. Click **Update** to save changes

Glossary

GlideRecord — ServiceNow's API for querying and manipulating database records

Business Rule — Server-side script that runs automatically on record events (insert, update, delete, query)

Before/After (Business Rule timing) — "Before" rules can modify or block a record before it saves; "After" rules run once the save is complete

Flow Designer — ServiceNow's modern, visual workflow automation builder

Data Pill — A drag-and-drop reference to a specific field's value, used inside Flow Designer to populate dynamic content

setAbortAction(true) — Script command used inside a Business Rule to cancel a save operation

gs.addErrorMessage() — Script command used to display an error message to the user

Dashboard Reference

Dashboard Name: Library Book Request Dashboard

- Combines two widgets pulling live data from the u_library_book_request table

Widgets:

- **Pie Chart** — Requests grouped by Status (Pending/Approved/Rejected)
- **List** — All requests with Request Number, Student Name, Book Title, Status

Purpose

Gives the librarian a single-screen, real-time view of request volume and details without manually filtering the table.

